

Shangyang Min

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EDUCATION

- Brown University** | Master of Science (Sc.M.) in Computer Science **09/2023 – 05/2025**
- GPA: 4.0/4.0
 - Artificial Intelligence/Machine Learning
 - Specialization: Multimodal, Brain-Inspired AI, Cross-domain Applications
- Michigan State University** | B.S. in Computer Science Engineering with High Honor **08/2019 – 05/2023**
- GPA: 3.95/4.0
 - Specialization: Software Development, Machine Learning, Computer Vision, Biomedical Imaging

SKILLS

- **Research Areas:** Deep Learning, Biomedical Engineering
- **Keywords:** CNNs, Transformers, Diffusion, Multimodal, Representation Learning, Radiomics
- **Additional Interests & Proficiencies:** LLM, HCI, Game Dev, Computational Neuroscience

Research

- Lee Lab** | Research Assistant **09/2024 – 05/2025**
- Investigated Brain-Computer-Interface integration with gaming, created a real-time EEG and Unity pipeline that enables natural avatar control.
 - Explored multimodal fusion techniques combining vision and EEG models for improved interaction fidelity.
- Human Augmentation and Artificial Intelligence Laboratory** | Research Assistant **05/2022 – 08/2023**
- Developed Feature Imitating Networks for biomedical imaging tasks, improving model robustness to limited annotations.
 - Led the project to an open-source release and a peer-reviewed publication; research supported by the National Science Foundation and the HFH+MSU Health Sciences Center Cancer Grant.
- Henry Ford Health System** | Research Assistant **09/2022 – 09/2023**
- Collaborated on a NIH-funded research program between Henry Ford Health System and Michigan State University.
 - Conducted machine learning analysis on tumor detection and radiomics features from DCE-MRI scans.

PROJECTS

- Game Development Studio** **09/2021 – 05/2023**
- Designed and Developed game mechanics and AI behaviors for multiple game projects
 - Worked under mentorship from Iron Galaxy Studio professionals, gaining professional development workflow experience.
- Selected Research Projects**
- **Trigger-Simulation Backdoor Removal Defense:** Proposed virtual-prompt realignment that slashed attack success rate from 93% to 0% backdoored Llama-2 variants within 7 epochs while keeping MMLU 42% accuracy.
 - **VQA with Action Insights:** Integrated external SOTA vision models and tokenized video-info annotations into GPT-4/Vicuna prompts, achieved high performance with zero-shot GPT-4 vs. fine-tuned SeViLa.
 - **See full list in my project webpage.** Recent submissions and projects, topics including multimodal, LLM, computer vision, computational neuroscience and HCI prototypes, etc.

PUBLICATIONS

- **S. Min, H. B. Ebadian, T. Alhanai and M. M. Ghassemi, Feature Imitating Networks Enhance the Performance, Reliability and Speed of Deep Learning on Biomedical Image Processing Tasks, 2024 46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC), Orlando, FL, USA, 2024, pp. 1-5, doi: 10.1109/EMBC53108.2024.10782373.**